

DATA ANALYSIS AND VISUALIZATION – LAB 12

Topic: Hands-On with D3.js (Advanced)

Q1 : Interactive Multi-Series Line Chart with Brush + Zoom

Dataset:

Columns: *date, temperature, humidity, wind*

Task Requirements:

1. Parse CSV; convert numeric columns (if required).
2. Plot 3 separate lines (temperature, humidity, wind) on the same chart.
3. Add:
 - **Brush** area at the bottom for selecting a date range.
 - **Zoom** so the main chart updates automatically when brush changes.
4. Add a **legend with clickable toggles**:
Clicking a legend item should hide/show that line (using update pattern).
5. Add **smooth transitions** when toggling series.

Q2: Force-Directed Graph With Drag + Highlighting + Node Grouping

Dataset: *nodes + links* JSON representing a collaboration network.

Task Requirements:

1. Create a force simulation using:
 - `forceManyBody`
 - `forceCenter`
 - `forceLink`

2. Add **drag behavior** (node must remain draggable).
3. When a node is hovered:
 - Highlight *all connected nodes + edges*.
 - Fade out all other nodes.
4. When clicked:
 - Fix the node's position (node.fx, node.fy).
 - Another click releases it.
5. Draw nodes with color **based on group membership** (e.g., department).

Q3: Treemap or Sunburst + Search + Zoom-in (Hierarchical Visualization)

Dataset: hierarchical JSON (e.g., software modules → files → LOC).

Task Requirements:

1. Load hierarchical JSON via `d3.hierarchy`.
2. Generate a **treemap or sunburst** layout.
3. Implement **click-to-zoom** into a node.
4. Add a **search bar**:
 - Typing a name highlights matching nodes (border + color flash).
 - Auto-zoom into the first match.
5. Display a breadcrumb trail showing the current zoom path.

Q4: Dynamic Update Pattern: Real-Time Dashboard Simulation

Dataset: A JS function that generates random data every 2 seconds:

```
function generateData() {  
  
  return Array.from({ length: 10 }, () => ({
```

```
id: crypto.randomUUID(),  
  
value: Math.random() * 100  
  
))) ;  
  
}
```

Task Requirements:

Create a mini dashboard containing **two synced charts**:

Chart A: Bar Chart

- Must use the **update pattern**:
 - Enter: new bars slide in from the right
 - Update: existing bars smoothly adjust height
 - Exit: removed bars shrink + fade out
- Bars must stay sorted by value (descending).

Chart B: Line Chart

- Appends a new point every update.
- Shifts the entire chart left (like a live ECG monitor).
- Implement a **clipping path** so the line does not overflow.

Dashboard Features

- Every 2 seconds, new data comes in → both charts update.
- Add a toggle button:
 - “Pause / Resume Streaming”
- Add a threshold slider:
 - Bars above threshold turn red.